Unit 5 Guided Notes — The Pythagorean Theorem

$a^2 + b^2 = c^2$. Add the legs to find the hypotenuse; subtract to find a leg.

Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

🗐 **Blank Worksheet (PDF)** ✅ **Answer Key (PDF)** 🗐 **Official PDF Packet** ♿ **EC/Modified Notes (PDF)**

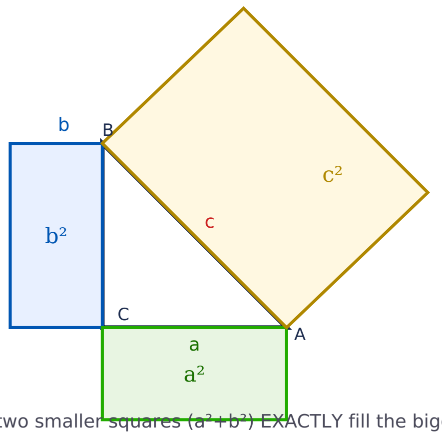
This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. $a^2 + b^2 = c^2$ — add the legs to find the hypotenuse; subtract to find a leg.

1 The Pythagorean Theorem

🔑 **THE THEOREM:** In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE: $a^2 + b^2 = c^2$.

a, b = legs (shorter sides) · c = hypotenuse (longest side, across from the right angle)

$a^2 + b^2 = c^2$



e two smaller squares (a²+b²) EXACTLY fill the biggest square (

⚠ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

Complete each blank on your printed copy, then click to check.

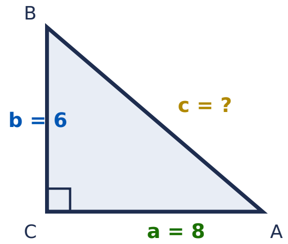
The two shorter sides of a right triangle are called the ______ (labeled a and b). The longest side, directly across from the right angle, is the ______ (labeled c).

The theorem written as an equation: $a^2 + b^2 =$ ______. It applies only to ______ triangles.

2 Finding the Hypotenuse

🔑 **WHEN TO USE:** When you know BOTH LEGS and need to find the HYPOTENUSE. Formula: $c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$

FIND THE HYPOTENUSE



Example: Legs are 6 and 8. Find the hypotenuse.

Step 1 — Write formula: $a^2 + b^2 = c^2$

Step 2 — Substitute: $6^2 + 8^2 = c^2$

Step 3 — Square: $36 + 64 = c^2$

Step 4 — Add: $100 = c^2$

✓ **Step 5 — Square root:** $c = \sqrt{100} = 10$

Practice — Find the Hypotenuse (c)

Legs 3 and 4: $3^2 + 4^2 = c^2 \rightarrow c = ?$

c =

Legs 6 and 8: $6^2 + 8^2 = c^2 \rightarrow c = ?$

c =

Legs 5 and 12: $5^2 + 12^2 = c^2 \rightarrow c = ?$

c =

Legs 8 and 15: $8^2 + 15^2 = c^2 \rightarrow c = ?$

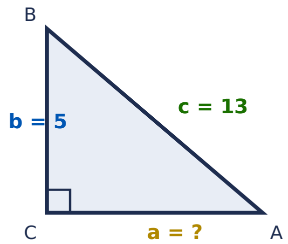
c =

3 Finding a Missing Leg

🔑 **WHEN TO USE:** When you know ONE LEG and the HYPOTENUSE. Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{c^2 - b^2}$

⚠ **REMEMBER:** Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



Example: One leg is 5, hypotenuse is 13. Find the other leg.

Step 1 – Write formula: $a^2 + b^2 = c^2$

Step 2 – Substitute: $5^2 + b^2 = 13^2$

Step 3 – Square: $25 + b^2 = 169$

Step 4 – Subtract 25 (finding a LEG = SUBTRACT): $b^2 = 144$

✓ **Step 5 – Square root:** $b = \sqrt{144} = 12$

ERROR ANALYSIS **A student finds a missing leg when c = 13, b = 5 by writing** $a^2 = 13^2 + 5^2 = 194$**. What's wrong?**

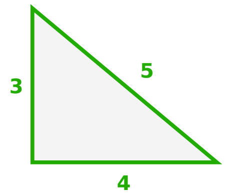
To find a leg you SUBTRACT: $a^2 = c^2 - b^2 = 169 - 25 = 144 \rightarrow a = 12$. Adding is only for the hypotenuse.

4 Converse of the Pythagorean Theorem

🔑 **THE CONVERSE:** Given three side lengths $a \le b \le c$, test: does $a^2 + b^2 = c^2$? If YES → right triangle. If NO → not a right triangle. Always test the two SMALLER sides against the LARGEST.

3, 4, 5

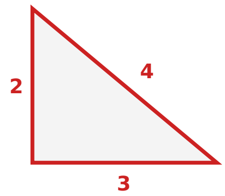
RIGHT triangle ✓



$4^2+3^2=25 = 5^2=25$

2, 3, 4

NOT a right triangle ✗



$3^2+2^2=13 \neq 4^2=16$

Quick Checks

MC **Sides 5, 12, 13 — right triangle?**

✓ $5^2 + 12^2 = 25 + 144 = 169$; $13^2 = 169$; $169 = 169$ → YES, right triangle.

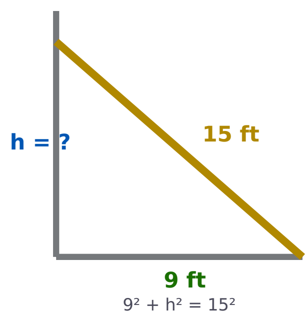
MC **Sides 6, 7, 10 — right triangle?**

✓ $6^2 + 7^2 = 36 + 49 = 85$; $10^2 = 100$; $85 \neq 100$ → NO, not a right triangle.

5 Word Problems

📌 **STRATEGY:** (1) Read carefully (2) Draw & label a diagram (3) Identify legs vs hypotenuse (4) Set up $a^2 + b^2 = c^2$ (5) Solve step by step.

LADDER AGAINST A WALL



Ladder Against a Wall: A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

Setup: $9^2 + h^2 = 15^2$

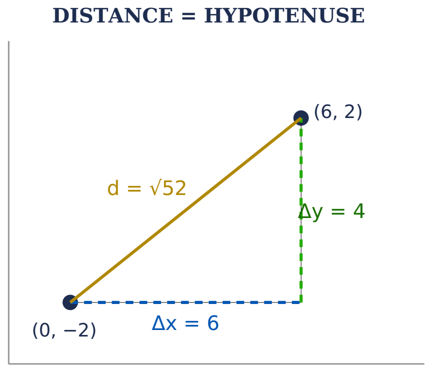
Square & subtract: $81 + h^2 = 225 \rightarrow h^2 = 144$

✓ **Square root:** $h = \sqrt{144} = 12$ ft

6 Distance on the Coordinate Plane

📐 **HOW IT WORKS:** Connect the two points to form a right triangle. $\Delta x = |x_2 - x_1|$ and $\Delta y = |y_2 - y_1|$. Then use $a^2 + b^2 = c^2$ where $a = \Delta x$, $b = \Delta y$, and c = distance.

⚠ **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!



Example: Distance from (0, −2) to (6, 2)

Δx: $|6 - 0| = 6$

Δy: $|2 - (-2)| = 4$

Setup: $\Delta x^2 + \Delta y^2 = d^2 \rightarrow 36 + 16 = 52$

✓ **Answer:** $d = \sqrt{52} \approx 7.21$

⚠ **WATCH YOUR SIGNS:** When subtracting negative coordinates, subtracting a negative means ADDING! Example: $5 - (-3) = 5 + 3 = 8$, NOT $5 - 3 = 2$.

✏ **Go to Unit 5 Practice →**

✏ Part 2 — Practice Problems

📌 **Show all work including the formula setup.**

A. Find the Hypotenuse

1) Legs 3, 4. c = ?

c =

2) Legs 5, 12. c = ?

c =

3) Legs 6, 8. c = ?

c =

4) Legs 8, 15. c = ?

c =

5) Legs 7, 24. c = ?

c =

6) Legs 9, 12. c = ?

c =

7) Legs 9, 40. c = ?

c =

8) Legs 20, 21. c = ?

c =

B. Find the Missing Leg

1) Known leg 4, hyp 5. Leg = ?

leg =

2) Known leg 12, hyp 13. Leg = ?

leg =

3) Known leg 8, hyp 10. Leg = ?

leg =

4) Known leg 15, hyp 17. Leg = ?

leg =

5) Known leg 9, hyp 15. Leg = ?

leg =

6) Known leg 24, hyp 25. Leg = ?

leg =

7) Known leg 20, hyp 29. Leg = ?

leg =

8) Known leg 16, hyp 20. Leg = ?

leg =

C. Word Problems

1. A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?

$h^2 + 5^2 = 13^2 \rightarrow h^2 = 169 - 25 = 144 \rightarrow h = 12$ ft.

2. A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?

$d^2 = 36^2 + 15^2 = 1296 + 225 = 1521 \rightarrow d = 39$ in.

3. A baseball diamond is a square with 90-ft sides. Distance from home to second base?

$d^2 = 90^2 + 90^2 = 16200 \rightarrow d \approx 127.3$ ft.

4. A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?

$h^2 + 60^2 = 100^2 \rightarrow h^2 = 10000 - 3600 = 6400 \rightarrow h = 80$ m.

5. A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?

$c^2 = 9^2 + 12^2 = 81 + 144 = 225 \rightarrow c = 15$ ft.

6. A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?

$d^2 = 24^2 + 10^2 = 576 + 100 = 676 \rightarrow d = 26$ m.

D. Distance on the Coordinate Plane

1) (1,2) → (4,6). d = ?

d =

2) (0,0) → (3,4). d = ?

d =

3) (−2,1) → (4,9). d = ?

d =

4) (1,−3) → (7,5). d = ?

d =

5) (−3,−1) → (5,5). d = ?

d =

6) (2,1) → (5,5). d = ?

d =

← Back to Unit 5